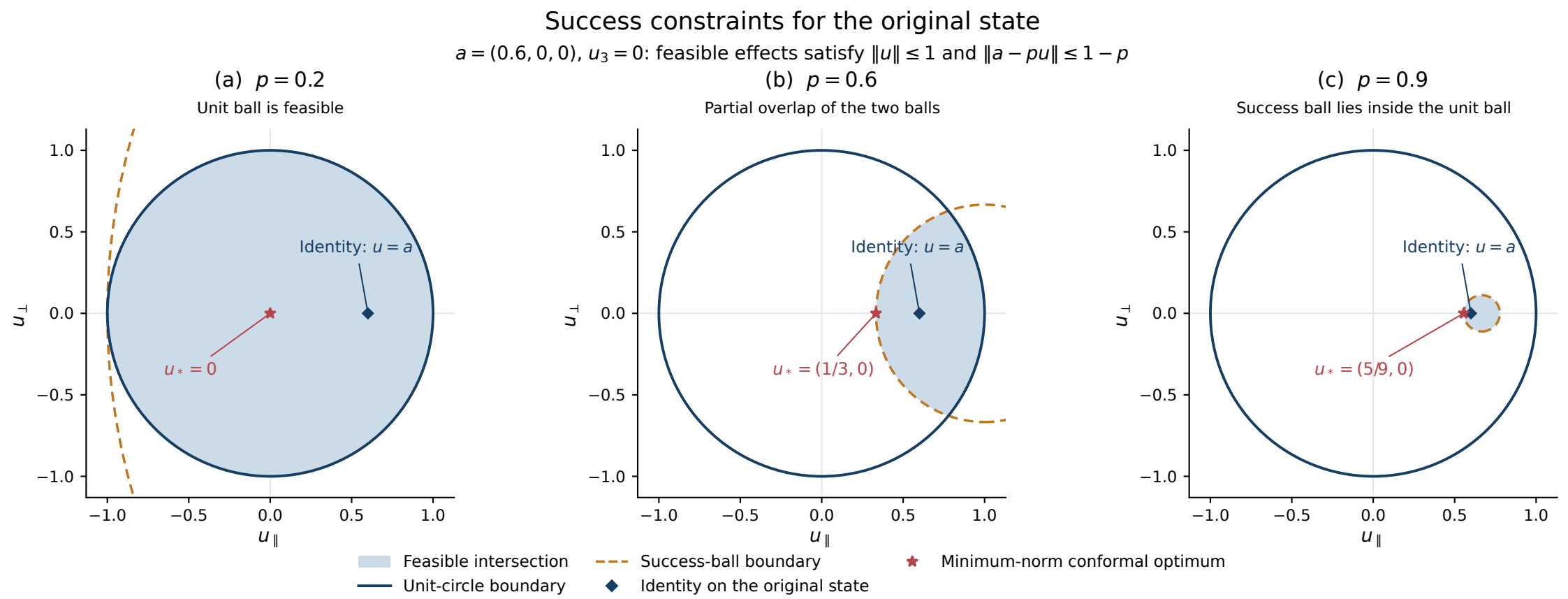
Success constraints for the original state
$a = (0.6, 0, 0)$, $u_3 = 0$: feasible effects satisfy $\|u\| \le 1$ and $\|a - pu\| \le 1 - p$
(a) $p = 0.2$
Unit ball is feasible
(b) $p = 0.6$
Partial overlap of the two balls
(c) $p = 0.9$
Success ball lies inside the unit ball
Identity: $u = a$
$u_* = 0$
$u_* = (1/3, 0)$
$u_* = (5/9, 0)$
$u_\perp$
$u_\parallel$
1.0
0.5
0.0
−0.5
−1.0
Feasible intersection
Unit-circle boundary
Success-ball boundary
Identity on the original state
Minimum-norm conformal optimum
p is the required success probability on the original input; u describes the effect after canonical whitening.